

TITLE: INTRODUCTION TO COMPUTER GRAPHICS & BASIC DRAWING USING C

INTRODUCTION

Computer graphics is a field that enables visual representation of data on a computer screen. In C programming, graphics can be implemented using the BGI (Borland Graphics Interface) or WinBGIm library. This lab focuses on understanding the basic structure of a graphics program in C, initializing graphics mode, drawing basic shapes, and using colors effectively. By the end of this experiment, students will gain hands-on experience with drawing pixels, lines, rectangles, and circles, and understanding the concept of a coordinate system in graphics.

Theory

Graphics Mode Initialization

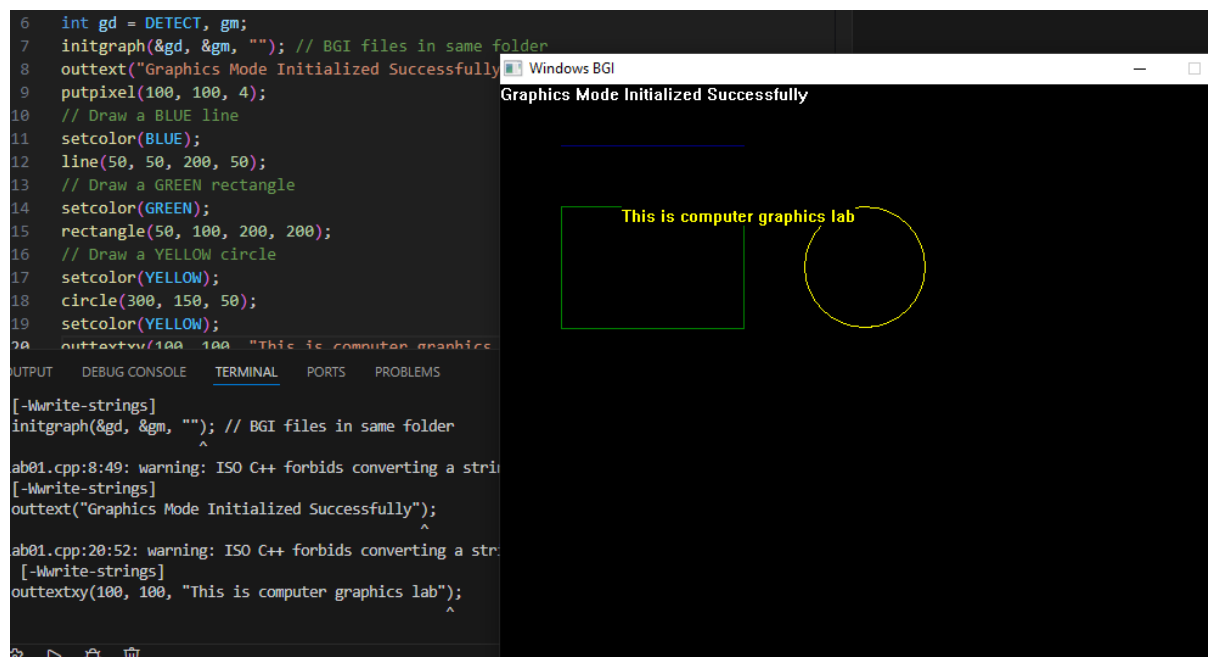
To work with graphics in C using BGI/WinBGIm libraries, the following steps are required:

- Initializing Graphics Driver – Prepares the system for graphics operations.
- Initializing Graphics Mode – Sets the screen mode for drawing.
- Specifying BGI File Directory – Ensures graphics drivers can access necessary files. The current folder can also be used.

Coordinate System in Graphics:

- Origin (0,0) is at the top-left corner of the screen.
- X-axis increases toward the right.
- Y-axis increases downward.

OUTPUT



The screenshot displays a C++ IDE with a code editor on the left and a graphical window on the right. The code in the editor initializes the BGI graphics mode, draws a blue line, a green rectangle, and a yellow circle, and prints text to the screen. The graphical window, titled 'Windows BGI', shows the output: a blue line, a green rectangle, a yellow circle, and the text 'This is computer graphics lab' in yellow.

```
6 int gd = DETECT, gm;  
7 initgraph(&gd, &gm, ""); // BGI files in same folder  
8 outtext("Graphics Mode Initialized Successfully");  
9 putpixel(100, 100, 4);  
10 // Draw a BLUE line  
11 setcolor(BLUE);  
12 line(50, 50, 200, 50);  
13 // Draw a GREEN rectangle  
14 setcolor(GREEN);  
15 rectangle(50, 100, 200, 200);  
16 // Draw a YELLOW circle  
17 setcolor(YELLOW);  
18 circle(300, 150, 50);  
19 setcolor(YELLOW);  
20 outtextxy(100, 100, "This is computer graphics lab");
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS PROBLEMS

```
[-Wwrite-strings]  
initgraph(&gd, &gm, ""); // BGI files in same folder  
ab01.cpp:8:49: warning: ISO C++ forbids converting a string  
[-Wwrite-strings]  
outtext("Graphics Mode Initialized Successfully");  
ab01.cpp:20:52: warning: ISO C++ forbids converting a string  
[-Wwrite-strings]  
outtextxy(100, 100, "This is computer graphics lab");
```

CONCLUSION

In this lab, we learned the fundamentals of computer graphics using C and the WinBGIm library. We successfully initialized the graphics mode, explored the coordinate system, and drew basic shapes such as pixels, lines, rectangles, and circles. We also learned how to use different colors for drawing and setting the background. This experiment provided hands-on experience in understanding how graphics programs are structured in C and how visual elements are represented on a screen. Overall, it strengthened our understanding of graphics programming concepts and prepared us for more advanced graphics applications.